Hertie School

[Thesis Title — TBD]

Master's Thesis submitted

in partial fulfillment of the requirements for the degree of

Master of Data Science for Public Policy at the Hertie School

Class of 2026

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Word Count: 4336

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3.1 Search and rescue as a guardrail at sea

International law places a duty on states and shipmasters to assist persons in distress at sea, regardless of nationality or legal status. The duty is codified in Article 98 of the UN Convention on the Law of the Sea (UNCLOS), which obliges both shipmasters to render assistance to any person in danger at sea and coastal states to maintain effective search-and-rescue services; its translation into EU-Mediterranean practice is analysed by Moreno-Lax (2018). In the Central Mediterranean this duty is carried out by a dispersed set of actors: state navies and coast guards, EU agencies, international organisations, merchant vessels, and NGO rescue ships. Their legal mandates differ, but the relevant operational output is the same: when a vessel becomes unable to complete the crossing safely, some actor can locate it, reach it, and prevent distress from becoming drowning.

The argument developed here is the *guardrail* view of SAR. Rough seas are the physical hazard. SAR does not remove that hazard, and it does not determine whether smugglers launch a boat on a given day. It changes what happens after a hazardous crossing has begun. When rescue capacity is close enough and willing enough to intervene, distress can become rescue, delay, or disembarkation. When that capacity is absent or redirected toward interception, the same physical hazard is more likely to become a recorded death. The theoretical object is therefore not the level of deaths before and after a policy change, but the gradient that maps sea state into mortality.

3.2 The 2017 shift: from rescue to interception

Before 2017, the Central Mediterranean contained substantial European and civil-society rescue capacity. The Italian state-led Mare Nostrum operation, EUNAVFOR MED Sophia, and NGO vessels operating close to the Libyan coast together formed the corridor's rescue infrastructure (Cusumano & Villa, 2021; Rodríguez Sánchez et al., 2023). This infrastructure was politically contested, but operationally meaningful: Cusumano & Villa (2021) documents that NGOs alone assisted approximately 120,000 migrants between 2014 and 2019.

The 2017 Italy–Libya Memorandum of Understanding (MoU) marks a qualitative shift in this infrastructure. The Minniti–Gentiloni Code of Conduct restricted NGO operations (Cusumano & Villa, 2021), subsequent Salvini decrees criminalised NGO activity (Cusumano & Bell, 2021; Punzo & Scaglione, 2024), and interview evidence describes a move toward a "pull-back regime" (Düvell & Denaro, 2024). At the same time, EU and Italian support increased the operational role of the Libyan Coast Guard, whose mandate is interception and return to Libya rather than disembarkation in a European place of safety. Moreno-Lax (2018) describes this as a “rescue-through-interdiction / rescue-without-protection” paradigm: the language of rescue remains, but the institutional content shifts from protection to containment.

This matters for the empirical design because the MoU is not a clean switch from SAR to no SAR. It is a bundled policy-regime change: NGO restrictions (Cusumano & Villa, 2021), Italian and EU operational changes (Punzo & Scaglione, 2024), Libyan interceptions, boat-composition responses (Deiana et al., 2024), and broader externalisation politics (Wolff, 2024) move together. The MoU is therefore a proxy for a change in the rescue/interception regime, not an exogenous treatment that isolates SAR alone. The empirical question must be phrased accordingly.

3.3 Why measuring SAR’s effect on mortality directly is hard

A direct estimate of how many lives SAR saved would require at least three objects: the counterfactual rescue capacity that would have existed absent the policy shift, the true number of deaths under each regime, and the true number of crossing attempts. None is observed cleanly. The data limitations are not incidental; they are part of the setting.

Departures are not observed. No public dataset records all departures from the North African shore. Frontex records events its operational assets engage with; coast-guard pullback series record some returns to Libya and Tunisia; death datasets record fatal outcomes; smugglers’ departures are not observed. The constructed crossing-attempt measure used later in the paper adds Frontex-recorded persons, LCG/TCG pullbacks, and IOM dead or missing persons. This is a useful lower-bound exposure measure, not observed daily departures.

Recorded deaths are a selected subset of true deaths. The empirical analysis uses UNITED as the primary mortality source and IOM Missing Migrants Project data as a comparison series. This choice is substantive. In the corridor sample used for the main models, UNITED records more total deaths but fewer death-days than IOM; the two series are strongly correlated at monthly frequency but differ in event coverage and verification practices. That divergence is useful triangulation, but it also shows why a single death series cannot be treated as complete; methodological reviews of Mediterranean mortality estimation reach the same conclusion at the corridor level (Kovras & Robins, 2016; Last & Spijkerboer, 2014).

The selection mechanism for what gets recorded changes when the policy regime changes. Before 2017, NGO presence was a major source of incident documentation; after 2017, NGO retreat and reclassification of operations altered what is seen and how it is categorised (Düvell & Denaro, 2024). Pullbacks also remove people from the at-sea population without necessarily generating Mediterranean mortality records. The parallel case of US-Mexico border deaths is instructive: undercounting of around 19% has been documented within a single jurisdiction, with rates fluctuating over time (Leutert, 2024). A before/after comparison of recorded deaths would therefore mix mortality, reporting, and exposure changes.

For this reason, the paper does not claim to estimate a complete fatality rate or a total number of lives saved. The primary empirical object is the daily association between sea state and recorded deaths. Constructed-exposure models are reported as sensitivity checks, and they enter exposure flexibly because the data do not support imposing a one-for-one proportional relationship between attempts and recorded deaths.

3.4 The wave–mortality slope as a sensible empirical strategy

The key empirical move is to study how the slope of recorded deaths with respect to lagged significant wave height (SWH) changes across policy regimes. SWH is useful because day-to-day marine weather is plausibly exogenous to the migration-control regime. The Italy–Libya MoU does not cause rough seas; rough seas do, however, affect both the decision to depart and the probability that a crossing becomes fatal.

The sign of the observed daily count slope is therefore theoretically ambiguous. Rough seas can reduce departures or delay crossings, pushing recorded deaths down on rough days. Conditional on a boat being at sea, the same rough seas increase the probability of distress and death. SAR capacity moderates the second channel: when rescue capacity is high, the lethality margin is partly buffered; when rescue is reduced or converted into interception, the lethality margin should become stronger. The guardrail prediction is therefore not simply "rough seas increase deaths." It is that the SWH-death gradient shifts upward when the rescue/interception regime weakens the guardrail.

This framing also explains why the count model is the main design. A true risk model would require observed departures. Since departures are not observed, the count model estimates the reduced-form gradient of recorded deaths on SWH. The rate-like model adds the constructed lower-bound exposure measure, but as a free log covariate rather than an offset. If exposure were a valid denominator with unit elasticity, an offset would be appropriate. The scripts explicitly test this proportionality and reject it, so the theory should not rest on a fatality-rate interpretation.

The remaining identifying concern is differential reporting. If missing deaths are a roughly constant share of true deaths within each weather regime, the slope comparison remains informative even though levels are biased. If, instead, post-2017 reporting worsens especially on rough days, the SWH gradient could partly reflect changing observation rather than changing lethality. The analysis responds by estimating the same models on UNITED and IOM, by checking future-weather placebo windows, and by using SAR-capacity, boat-composition, crossing-volume, and period decompositions as diagnostic evidence rather than as separate causal designs.

3.5 Competing predictions

The literature implies four competing readings of SAR, each with a distinct implication for the SWH-death gradient.

The *guardrail* view holds that SAR moderates the conditional translation of maritime hazard into death once a boat is at sea. Its prediction is sharp: the SWH-mortality gradient should shift upward when rescue capacity is reduced, and higher measured SAR capacity should flatten the gradient at any given regime.

The *pull-factor* view holds that SAR primarily affects the volume of attempts rather than conditional lethality (Rodríguez Sánchez et al., 2023). If this channel dominates, any post-MoU change in the SWH gradient should operate through changes in crossing volume and be absorbed once exposure is controlled.

The *moral-hazard* view holds that SAR availability changes the composition of attempts, for instance toward flimsier boats and willingness to depart on rougher days (Deiana et al., 2024). If this channel dominates, the SWH gradient should be steeper when SAR is abundant, or should be substantially attenuated by boat-composition controls after the MoU.

The *deterrence-and-diversion* view holds that the post-2017 regime deterred or rerouted movement on the Central Mediterranean route (Camarena et al., 2020; Hoffmann Pham & Komiyama, 2024; Tafani & Riccaboni, 2025). If this channel dominates, observed mortality may fall because the denominator falls or shifts to other routes, and the weather gradient need not identify rescue capacity at all.

These four mechanisms operate at different *stages* of the journey: selection into crossing (moral hazard, pull factor), route or destination (diversion), and outcome conditional on crossing (guardrail). This staging is what makes the predictions partly separable. The guardrail view alone predicts that, conditional on exposure and boat composition, the SWH-mortality slope should respond to the policy regime; the other views do not.

The evidence required to sustain the guardrail interpretation is therefore not one coefficient but a pattern: an upward post-MoU shift in the SWH-death gradient across death sources and count families; persistence under exposure and boat-composition checks; absence of the same pattern in future-weather placebos; and a negative interaction between SWH and direct SAR-capacity measures. The spatial analogue of this prediction has already been documented on the same policy event: Zambiasi & Albarosa (2025) find increased deadly-incident probability in Libyan-Coast-Guard-patrolled areas post-MoU, and Prieto-Flores (2025) report a parallel pattern on the Western Mediterranean route after the 2018 EU-Morocco agreements. This thesis examines whether the same prediction holds in time.

3.6 Theoretical estimand

Let D_t denote recorded deaths on day t, s_t lagged SWH, A_t true crossing attempts, R_t SAR capacity, and q_t the probability that deaths are recorded. A stylised representation is

$$\mathbb{E}[D_t \mid s_t, R_t] \;=\; q(s_t, R_t)\, A(s_t, R_t)\, m(s_t, R_t),$$

where $m(s_t, R_t)$ is the latent mortality risk per attempt. The guardrail claim concerns m: higher SAR capacity should reduce the marginal effect of rough seas on lethality. Formally,

$$\Delta_m \;=\; \left.\frac{\partial m(s, R)}{\partial s}\right|_{R=R_{\text{low}}} \;-\; \left.\frac{\partial m(s, R)}{\partial s}\right|_{R=R_{\text{high}}} \;>\; 0.$$

This is the conceptual estimand. It is not directly observed because A_t, q_t, and $m(s_t, R_t)$ are not separately identified in the available data. The observable object is the reduced-form slope of recorded deaths on SWH, which decomposes as

$$\frac{\partial \log \mathbb{E}[D_t \mid s_t, R_t]}{\partial s_t} \;=\; \underbrace{\frac{\partial \log q}{\partial s_t}}_{\text{recording}} + \underbrace{\frac{\partial \log A}{\partial s_t}}_{\text{exposure}} + \underbrace{\frac{\partial \log m}{\partial s_t}}_{\text{lethality}}.$$

The guardrail prediction concerns the cross-derivative of the third term with respect to SAR capacity: $\partial^2 \log m/\partial s\, \partial R < 0$. Identifying this cross-derivative from the reduced-form slope shift across regimes requires assumptions on the first two channels. Specifically, either (i) $\partial^2 \log q/\partial s\, \partial R$ and $\partial^2 \log A/\partial s\, \partial R$ are zero, meaning recording and exposure sensitivities to weather do not change with the rescue/interception regime; or (ii) they move in the opposite direction to the lethality channel, in which case the recovered slope shift is a lower bound on the lethality shift.

The empirical strategy assesses these assumptions but cannot rule them out fully. The future-weather placebos and the IOM/UNITED cross-source comparison address differential recording; the constructed-exposure sensitivity and the boat-composition controls address the exposure channel. The observable implication is therefore weaker than the structural cross-derivative: if the guardrail channel is important and the auxiliary assumptions hold at least directionally, the SWH gradient in recorded deaths should become less negative or more positive when SAR capacity is reduced, and it should flatten when measured SAR capacity is higher.

3.7 From theoretical to empirical estimand

The main empirical estimand is the change in the SWH-recorded-death slope around the post-MoU regime:

$$\beta_3 = \beta_{\text{post}} - \beta_{\text{pre}}$$

where β_{pre} and β_{post} are the slopes of daily recorded deaths on the lagged five-day mean of SWH before and after 1 July 2017. UNITED is the primary mortality series; IOM is an independent comparison series. Negative pre-period slopes are not a problem for the theory: they are consistent with rough seas suppressing departures when the guardrail is stronger. The guardrail prediction is an upward shift in the slope after the policy-regime change.

The interpretation of β_3 requires caution. It is a reduced-form policy-regime moderation parameter, not an isolated SAR effect. The MoU bundles NGO restrictions, Italian decrees, Libyan interceptions, route and boat-composition changes, push factors, and possible changes in reporting. The mechanism analysis therefore replaces the binary post-MoU indicator with continuous SAR-capacity moderators, but those moderators are themselves endogenous to operational decisions and distress events. The period and rolling-gradient analyses likewise treat the regime change as gradual and uneven, not as a permanent step function.

A note on interpretation: mechanically, β_3 is the interaction between a continuous regressor (lagged SWH) and a time indicator (post-MoU), which is algebraically the same object as a continuous-regressor difference-in-differences estimator. The interpretation here is more limited because the corridor has no second unit to serve as a control: β_3 does not identify a counterfactual untreated trajectory. What it identifies is a within-corridor change in the conditional slope of recorded deaths on SWH, given month-year fixed effects. The "policy-regime moderation" label is meant to preserve this distinction.

The empirical claim this thesis can sustain is therefore precise: recorded deaths became more sensitive to lagged rough seas after the rescue/interception regime changed, and the broader pattern of robustness and mechanism evidence is more consistent with the guardrail view than with pure pull-factor, composition-only, or diversion-only explanations. The evidence does not by itself identify the number of deaths caused by SAR withdrawal.

3.8 Research question

This framing yields a single empirical question:

> How does the marginal association between sea state and recorded deaths change across SAR/policy regimes?

A positive shift is consistent with the guardrail view. A null shift would fit a pure volume or pull-factor account. A shift in the opposite direction would fit a moral-hazard account in which abundant SAR makes crossings more weather-sensitive. Because the data cannot isolate all channels, the results section evaluates the whole pattern of evidence rather than treating one coefficient as dispositive.

4 Methods and Data

4.1 Setting and unit of observation

The analysis is a daily panel of the Central Mediterranean Route (CMR). The unit of observation in the main models is a corridor-day. This level is useful for two reasons. First, the theoretical mechanism is short-run: sea state changes from day to day and can affect both departure decisions and the survival margin once a boat is at sea. Second, the closest methodological precedent in this literature uses daily variation in significant wave height (SWH) to study behavioural responses to SAR operations in the same corridor (Deiana et al., 2024). Aggregating to months would remove much of the weather variation that identifies the slope of interest.

The Frontex-bounded panel covers 1 January 2014 to 31 May 2023, for 3,438 calendar days. The end date is not chosen by hand. It is read from the daily LCG/TCG interception series used in the crossing-exposure construction, which currently ends on 31 May 2023. The main regression sample starts on 15 January 2014, after lagged crossing and SWH windows are available, and runs through 31 May 2023. This gives 3,424 available corridor-days before fixed-effect cells with no outcome variation are dropped by the count estimators.

The policy-regime indicator is defined as $Post_t = 1\{t \geq 1 \text{ July } 2017\}$. This date follows the operational definition used in the main model scripts. It is not interpreted as a sharp causal switch. Rather, it marks the post-MoU rescue/interception regime for the reduced-form comparison. Additional period and rolling-window analyses relax the two-period structure and examine whether the weather-mortality gradient evolves gradually over later operational regimes.

4.2 Data sources and construction

The empirical panel combines seven data sources: ERA5 wave data, UNITED for Intercultural Action deaths, IOM Missing Migrants Project deaths, Frontex Themis incident records, LCG/TCG pullback counts, UNHCR arrivals, and ACLED conflict data. The construction is automated in the project scripts. The daily panel used by the models is saved as `analysis/data/daily_panel_complete.RDS`; source-specific mortality series are then rebuilt in the analysis scripts from the raw processed death files so that the outcome filters are explicit and consistent.

The weather exposure is constructed from ERA5 daily netCDF files for 2008-2025. The cleaning script aggregates significant wave height over the core corridor polygon saved in `data/processed/core_corridor.RDS`. The resulting daily SWH series is a simple spatial mean over ocean grid cells in the corridor. The same corridor polygon is used later to spatially filter death records, so the weather and mortality measures refer to the same sea area. The primary regressor is the lagged five-day mean of SWH, $SWH_{t-1:t-5}$. The script also constructs one-day, three-day, and seven-day lagged means. All lagged windows exclude day-t weather, which keeps the regressor temporally prior to the recorded outcome day.

Mortality is measured with two independent event datasets. UNITED is the primary outcome series. The UNITED builder keeps deaths in Algeria, Italy, Libya, Malta, Tunisia, and open-sea records coded as Mediterranean; restricts causes to drowning or other/unknown maritime deaths; drops records without usable coordinates when the spatial filter is applied; and spatially joins the remaining incidents to the core corridor polygon. IOM is estimated separately as a comparison series. The IOM builder keeps Central Mediterranean incidents in Algeria, Italy, Libya, Malta, and Tunisia; restricts the analytical outcome to incident records rather than split incidents; keeps drowning and mixed or unknown causes; and applies the same corridor-polygon spatial filter. Split incidents are excluded from the analytical IOM outcome because they spread one reported event across multiple calendar dates, weakening the link between recent sea state and the event date and reducing comparability with UNITED.

In the 2014-2023 corridor window, the matched analytical filters produce 20,368 UNITED deaths on 503 death-days and 15,285 IOM dead or missing on 616 death-days. The two sources do not agree event by event. UNITED records more deaths but on fewer death-days, while IOM records deaths on more days. This is why the paper treats UNITED as the primary source and IOM as a triangulation check rather than pooling the two series. Their monthly counts are highly correlated in the constructed corridor sample, but the daily differences are substantively meaningful and remain part of the measurement uncertainty.

Frontex Themis records provide daily information on CMR events observed by Frontex-linked operations. The panel keeps departures from Libya, Tunisia, and Algeria and aggregates events, persons, SAR status, interceptor type, detection method, departure country, and boat type to the day. SAR status in this source refers to the European-side engagement of a detected boat. It is therefore kept conceptually separate from LCG/TCG pullbacks, which are returns to Libya or Tunisia rather than rescues to a European place of safety.

The constructed exposure measure is a lower-bound count of observed crossing attempts:

$$C_t = \text{Frontex persons}_t + \text{LCG/TCG pullbacks}_t + \text{IOM dead or missing}_t.$$

This follows the broad logic in the CMR migration-flow literature of combining arrivals/interceptions, pushbacks, and deaths to approximate attempted crossings (Deiana et al., 2024; Rodríguez Sánchez et al., 2023). In this thesis, the measure is daily and deliberately conservative. It excludes undetected arrivals computed from UNHCR-Frontex daily gaps because the daily timing mismatch between event-day and landing-day records inflates that gap mechanically. The scripts retain UNHCR arrivals for validation, but the regression denominator is `frx_persons + lcg_tcg_pushbacks + n_dead_missing`, where `n_dead_missing` is the broad IOM count in the base panel rather than a source-specific analytical outcome. Across the full panel this measure sums to 1,082,372 observed attempts, consisting of 819,576 Frontex-recorded persons, 242,934 LCG/TCG pullbacks, and 19,862 broad IOM dead or missing persons.

LCG and TCG pullbacks enter the daily panel through a Denton-disaggregated series. Monthly pullback counts are allocated to days using Frontex daily departures as the high-frequency indicator, then rounded so monthly totals are preserved. ACLED data are used for push-factor checks. The conflict variable used in those checks is a Libya composite of battles, explosions/remote violence, and violence against civilians, lagged by one week in the decomposition script. These variables are not part of the primary count specification; they are used to examine whether the SWH-mortality shift is better explained by changing conflict pressure.

4.3 Outcome, exposure, and estimand

The main outcome is the daily count of recorded deaths in source j, D_t^j, where $j \in \{\mathrm{UNITED}, \mathrm{IOM}\}$. The estimand is the change in the marginal association between lagged sea state and recorded deaths across the post-MoU rescue/interception regime. It is not the level change in deaths, not a completed-crossing fatality rate, and not the number of lives saved by SAR. The object of interest is the slope:

$$\beta_3 = \frac{\partial \mathbb{E}[D_t^j \mid SWH_t, Post_t = 1]}{\partial SWH_t} - \frac{\partial \mathbb{E}[D_t^j \mid SWH_t, Post_t = 0]}{\partial SWH_t}.$$

This estimand matches the theory section. If SAR acts as a guardrail, rough seas should translate into recorded deaths more strongly when rescue capacity is reduced or redirected toward interception. A positive β_3 is therefore consistent with the guardrail interpretation. It is not sufficient by itself to prove that SAR withdrawal caused deaths, because the post-MoU regime bundles many operational and reporting changes.

4.4 Main count specification

The primary specification estimates daily recorded deaths as a function of lagged SWH and its interaction with the post-MoU indicator:

$$\mathbb{E}[D_t^j \mid SWH_t, Post_t, \alpha_{m(t)}] = \exp\left(\alpha_{m(t)} + \beta_1 SWH_{t-1:t-5} + \beta_3 SWH_{t-1:t-5} \times Post_t\right).$$

The fixed effect $\alpha_{m(t)}$ is a month-year fixed effect. It absorbs seasonality and common monthly shocks; the model does not estimate a separate post-MoU level shift. With this fixed-effect structure, identification comes from within-month variation in lagged sea state and from how the SWH slope differs before and after the post-MoU date. The coefficient

β_1 is the pre-period SWH slope, while β_3 is the post-MoU change in that slope. Results are reported as β_1, β_3, and the implied post-period slope $\beta_1 + \beta_3$, with a delta-method standard error for the latter.

The count model is estimated with both negative binomial maximum likelihood and Poisson quasi-maximum likelihood. The negative binomial model is useful because recorded deaths are overdispersed and heavily zero-valued. The Poisson QMLE is reported alongside it because it estimates the conditional mean directly and does not require equality of the conditional variance and mean for consistency of the mean parameters. UNITED is the primary source in the presentation; IOM is estimated with the same specification and sample logic as an independent comparison.

Inference uses Newey-West standard errors with a 14-day lag and `panel.id = ~ unit + date` in the daily corridor models. The 14-day bandwidth allows for serial correlation over a period longer than the five-day SWH window and the lagged crossing windows used in robustness checks. The scripts also report standard errors clustered by month-year as an alternative inference choice. The main conclusions are evaluated against this broader pattern, not against a single standard-error convention.

4.5 Rate-like sensitivity with constructed exposure

Because true departures are unobserved, the count specification is the main design. A risk or fatality-rate model would require a denominator that measures all attempts. The constructed exposure measure C_t is informative, but it is a lower bound and may be measured with error. For this reason, the rate-like models do not impose an offset. Instead, they estimate

$$\mathbb{E}[D_t^j \mid SWH_t, Post_t, C_t, \alpha_{m(t)}] = \exp\left(\alpha_{m(t)} + \beta_1 SWH_{t-1:t-5} + \beta_3 SWH_{t-1:t-5} \times Post_t + \gamma \log(C_t)\right),$$

on days with $C_t > 0$. This keeps exposure as a flexible covariate rather than forcing deaths to scale one-for-one with observed attempts. The distinction is empirically important. In the current output, the coefficient on `log(crossing_attempts)` is far below one for both UNITED and IOM, and Wald tests reject the unit-elasticity restriction. The rate-like models should therefore be read as volume-controlled count models, not as estimates of a true daily fatality rate. The shared rate-like sample contains 2,125 days with positive constructed exposure before all-zero fixed-effect cells are dropped.

4.6 Identification assumptions and threats

The identifying assumption for the reduced-form slope is that, conditional on month-year fixed effects, day-to-day variation in lagged SWH is as-good-as random with respect to other determinants of recorded mortality. This assumption is most plausible for marine weather itself: the post-MoU policy regime does not cause rough seas. The stronger interpretation, that the post-MoU change in the SWH-death slope captures the changed rescue/interception regime, additionally requires that no other post-2017 factor changed the weather-to-death gradient in the same direction.

That stronger assumption is demanding. The MoU coincided with NGO restrictions, Italian decrees, expanded Libyan interceptions, changes in smuggler strategy, changing boat composition, conflict dynamics in Libya, and potential changes in reporting. The paper therefore treats the post-MoU interaction as a reduced-form moderation parameter. It asks whether the weather-mortality slope changed around the policy regime, then examines whether the surrounding diagnostics are more consistent with the guardrail mechanism than with pure volume, composition, or diversion explanations.

Measurement error is the second major threat. Recorded deaths are not true deaths, and observed attempts are not true departures. Differential reporting would be especially problematic if post-2017 deaths became more likely to be missed on rough-weather days. The two-source design partly addresses this: UNITED and IOM use different collection and verification practices, so a similar pattern across both sources is less likely to be a single-source artefact. This does not remove reporting bias, but it makes the measurement assumption visible rather than hidden.

4.7 Mechanism and robustness specifications

The mechanism analysis replaces the binary policy-regime interaction with continuous SAR-capacity moderators. The main mechanism specification is

$$\mathbb{E}[D_t^j] = \exp\left(\alpha_{f(t)} + \theta_1 SWH_{t-1:t-5} + \theta_2 R_{t-1:t-7} + \theta_3 SWH_{t-1:t-5} \times R_{t-1:t-7}\right),$$

where $R_{t-1:t-7}$ is a weekly lagged SAR measure. The scripts use two versions: the share of Frontex incidents classified as SAR over the previous week, and the log of one plus the number of persons in Frontex SAR events over the previous week. Both moderators are standardised. The share measure is close to the theoretical idea of SAR prevalence, but it has a denominator concern when non-SAR events and pullbacks change sharply. The rescued-persons measure is therefore reported as a complementary absolute-capacity proxy. These models are mechanism checks, not separate identification strategies, because measured SAR capacity is itself endogenous to crossings, distress, detection, and operational decisions.

Robustness checks are organised around the main threats to interpretation. Exposure-window checks re-estimate the SWH-post-MoU model using lagged one-day, three-day, five-day, and seven-day SWH windows, and compare them to future-weather placebo windows. Fixed-effect and inference checks vary the calendar controls, Newey-West bandwidths, and clustering. Lagged crossing controls add pre-determined crossing-volume measures.

Boat-composition checks restrict to days with Frontex-observed boats and add inflatable and wooden boat shares, plus an interaction between SWH and inflatable share. ACLED checks test whether the post-MoU SWH-death shift varies with lagged Libya conflict. Period and rolling-gradient models relax the binary post-MoU structure, while zone-level models examine whether the pattern differs across broad African and European SAR blocs. These checks do not turn the design into a clean causal experiment, but they discipline the interpretation of the main slope estimate.

5 Analysis and Results

6 Conclusion and Further Work

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Appendix